problem statement

Proposition 0.1. *Define $g_{n,i}(x) = I_x(i, n-i+1)$, we want to prove $g^{-1}_{n,n-j}(x)^n$ is an increasing function with respect to n, or $g_{n,n-j}(x^{\frac{1}{n}})$ is a decreasing function with respect to n. We have known $g_{n,n-j}(x)$ is a decreasing function with respect to n and $g^{-1}_{n,n-j}(x)$ is an increasing function with respect to n.*

I'm going to drop the $g_{i,n}$ notation here because it is very confusing. That notation should only be used when there are space constraints.

We will show that

$$I_{x^{\frac{1}{n}}}(n-j, j+1) - I_{x^{\frac{1}{n+1}}}(n-j+1, j+1) > 0.$$

Assume $x^{\frac{1}{n}} > \frac{n-j}{n}$:

Integration by parts allows us to write

$$\begin{aligned}
&I_{x^{\frac{1}{n}}}(n-j, j+1) - I_{x^{\frac{1}{n+1}}}(n-j+1, j+1) \\
&\quad = I_{x^{\frac{1}{n}}}(n-j+1, j+1) - I_{x^{\frac{1}{n+1}}}(n-j+1, j+1) + \frac{\left(x^{\frac{1}{n}}\right)^{n-j}\left(1-x^{\frac{1}{n}}\right)^{j+1}}{(n-j)B(n-j, j+1)} \\
&\quad = -\int_{x^{\frac{1}{n}}}^{x^{\frac{1}{n+1}}} \frac{y^{n-j}(1-y)^j}{B(n-j+1, j+1)} dy + \frac{\left(x^{\frac{1}{n}}\right)^{n-j}\left(1-x^{\frac{1}{n}}\right)^{j+1}}{(n-j)B(n-j, j+1)}.
\end{aligned}$$

The function $y^a(1-y)^b$ achieves its maximum value at $y^* = a/(a+b)$. The assumption that $x^{\frac{1}{n}} > \frac{n-j}{n}$ implies that the function $y^{n-j}(1-y)^j$ is decreasing over $(x^{\frac{1}{n}}, x^{\frac{1}{n+1}})$. This implies that

$$\begin{aligned}
&-\int_{x^{\frac{1}{n}}}^{x^{\frac{1}{n+1}}} \frac{y^{n-j}(1-y)^j}{B(n-j+1, j+1)} dy + \frac{\left(x^{\frac{1}{n}}\right)^{n-j}\left(1-x^{\frac{1}{n}}\right)^{j+1}}{(n-j)B(n-j, j+1)} \\
&\qquad \geq \frac{\left(x^{\frac{1}{n}}\right)^{n-j}\left(1-x^{\frac{1}{n}}\right)^{j}(x^{\frac{1}{n}} - x^{\frac{1}{n+1}})}{B(n-j+1, j+1)} + \frac{\left(x^{\frac{1}{n}}\right)^{n-j}\left(1-x^{\frac{1}{n}}\right)^{j+1}}{(n-j)B(n-j, j+1)}.
\end{aligned}$$

And $B(n-j+1,j+1) = B(n-j,j+1)\frac{n-j}{n+1}$ allows us to express the right hand side of the above inequality as

$$\begin{aligned}
&\frac{\left(x^{\frac{1}{n}}\right)^{n-j}\left(1-x^{\frac{1}{n}}\right)^{j}(x^{\frac{1}{n}}-x^{\frac{1}{n+1}})}{B(n-j+1,j+1)} + \frac{\left(x^{\frac{1}{n}}\right)^{n-j}\left(1-x^{\frac{1}{n}}\right)^{j+1}}{(n-j)B(n-j,j+1)} \\
&= \frac{\left(x^{\frac{1}{n}}\right)^{n-j}\left(1-x^{\frac{1}{n}}\right)^{j}(x^{\frac{1}{n}}-x^{\frac{1}{n+1}})}{B(n-j,j+1)\frac{n-j}{n+1}} + \frac{\left(x^{\frac{1}{n}}\right)^{n-j}\left(1-x^{\frac{1}{n}}\right)^{j+1}}{(n-j)B(n-j,j+1)} \\
&= \frac{\left(x^{\frac{1}{n}}\right)^{n-j}\left(1-x^{\frac{1}{n}}\right)^{j}}{B(n-j,j+1)(n-j)}\left((n+1)\left(x^{\frac{1}{n}}-x^{\frac{1}{n+1}}\right)+\left(1-x^{\frac{1}{n}}\right)\right) \\
&= \frac{\left(x^{\frac{1}{n}}\right)^{n-j}\left(1-x^{\frac{1}{n}}\right)^{j}}{(n-j)B(n-j,j+1)}\left(nx^{\frac{1}{n}}-(n+1)x^{\frac{1}{n+1}}+1\right)
\end{aligned}$$

Let $f_n(x) = nx^{\frac{1}{n}} - (n+1)x^{\frac{1}{n+1}} + 1$. Then

$$f_n^{'}(x) = x^{\frac{1-n}{n}} - x^{-\frac{n}{n+1}} = x^{\frac{1-n}{n}}(1-x^{-\frac{1}{n(n+1)}}).$$

Thus, $f_n^{'}(x) \leq 0$ for all $x \in (0,1)$ which implies that $f_n(x) \geq 0$ for all $x \in (0,1)$. Therefore, when $x^{\frac{1}{n}} > \frac{n-j}{n}$ we can conclude that

$$I_{x^{\frac{1}{n}}}(n-j,j+1) - I_{x^{\frac{1}{n+1}}}(n-j+1,j+1) \geq 0.$$

Assume $x^{\frac{1}{n+1}} < \frac{n-j}{n}$:

Integration by parts allows us to write

$$\begin{aligned}
&I_{x^{\frac{1}{n}}}(n-j,j+1) - I_{x^{\frac{1}{n+1}}}(n-j+1,j+1) \\
&= I_{x^{\frac{1}{n}}}(n-j,j+1) - \left(I_{x^{\frac{1}{n+1}}}(n-j,j+1) - \frac{\left(x^{\frac{1}{n+1}}\right)^{n-j}\left(1-x^{\frac{1}{n+1}}\right)^{j+1}}{(n-j)B(n-j,j+1)}\right) \\
&= \frac{\left(x^{\frac{1}{n+1}}\right)^{n-j}\left(1-x^{\frac{1}{n+1}}\right)^{j+1}}{(n-j)B(n-j,j+1)} + I_{x^{\frac{1}{n}}}(n-j,j+1) - I_{x^{\frac{1}{n+1}}}(n-j,j+1) \\
&= \frac{\left(x^{\frac{1}{n+1}}\right)^{n-j}\left(1-x^{\frac{1}{n+1}}\right)^{j+1}}{(n-j)B(n-j,j+1)} - \int_{x^{\frac{1}{n}}}^{x^{\frac{1}{n+1}}} \frac{y^{n-j-1}(1-y)^{j}}{B(n-j,j+1)}dy \\
&\geq \frac{\left(x^{\frac{1}{n+1}}\right)^{n-j}\left(1-x^{\frac{1}{n+1}}\right)^{j+1}}{(n-j)B(n-j,j+1)} - \frac{\left(x^{\frac{1}{n+1}}\right)^{n-j-1}\left(1-x^{\frac{1}{n+1}}\right)^{j}}{B(n-j,j+1)}\left(x^{\frac{1}{n+1}}-x^{\frac{1}{n}}\right) \\
&= \frac{\left(x^{\frac{1}{n+1}}\right)^{n-j-1}\left(1-x^{\frac{1}{n+1}}\right)^{j}}{(n-j)B(n-j,j+1)}\left(x^{\frac{1}{n+1}}(1-x^{\frac{1}{n+1}}) - (n-j)(x^{\frac{1}{n+1}}-x^{\frac{1}{n}})\right)
\end{aligned}$$

$$= \frac{\left(x^{\frac{1}{n+1}}\right)^{n-j-1}\left(1-x^{\frac{1}{n+1}}\right)^{j}}{(n-j)B(n-j,j+1)}\left((n-j)x^{\frac{1}{n}}-(n-j-1)x^{\frac{1}{n+1}}-x^{\frac{2}{n+1}}\right)$$

Now,

$$\begin{aligned} f'_{n,j}(x) &= \frac{(n+1)(n-j)}{j+1}\left(\frac{x^{\frac{1-n}{n}}}{n} - \frac{x^{-\frac{n}{n+1}}}{n+1}\right) - \frac{x^{\frac{1-n}{n}}}{n} \\ &= \frac{x^{\frac{1-n}{n}}}{n}\left(\frac{(n+1)(n-j)}{j+1}\left(1 - \frac{n}{n+1}x^{-\frac{n}{n+1}-\frac{1-n}{n}}\right) - 1\right) \\ &= \frac{x^{\frac{1-n}{n}}}{n}\left(\frac{(n+1)(n-j)}{j+1}\left(1 - \frac{n}{n+1}x^{-\frac{1}{n(n+1)}}\right) - 1\right). \end{aligned}$$

From here we see that there exists a single root,

$$\begin{aligned} &\frac{x^{\frac{1-n}{n}}}{n}\left(\frac{(n+1)(n-j)}{j+1}\left(1 - \frac{n}{n+1}x^{-\frac{1}{n(n+1)}}\right) - 1\right) \overset{\text{set}}{=} 0, \\ &\implies \frac{(n+1)(n-j)}{j+1}\left(1 - \frac{n}{n+1}x^{-\frac{1}{n(n+1)}}\right) = 1, \\ &\implies \frac{n}{n+1}x^{-\frac{1}{n(n+1)}} = 1 - \frac{j+1}{(n+1)(n-j)}, \\ &\implies x^* = \left(\frac{n+1}{n}\left(1 - \frac{j+1}{(n+1)(n-j)}\right)\right)^{-n(n+1)}. \end{aligned}$$

Simplifying further yields

$$\begin{aligned} \left(\frac{n+1}{n}\left(1 - \frac{j+1}{(n+1)(n-j)}\right)\right)^{-n(n+1)} &= \left(\frac{n+1}{n} - \frac{j+1}{n(n-j)}\right)^{-n(n+1)} \\ &= \left(\frac{(n+1)(n-j)}{n(n-j)} - \frac{j+1}{n(n-j)}\right)^{-n(n+1)} \\ &= \left(\frac{n(n-j)+n-2j+1}{n(n-j)}\right)^{-n(n+1)}. \end{aligned}$$

We have $x^* \geq 1$ whenever $n - 2j + 1 \leq 0$. We know that $f_{n,j}(0) = 1$ and $f_{n,j}(1) = 0$, and this implies that $f(x) > 0$ for all $x \in (0, 1)$. Now consider when $n - 2j + 1 > 0$. We can rewrite (**??**) as

$$f_{n,j}(x) = \frac{(n+1)(n-1)-(j+1)}{j+1}x^{\frac{1}{n}} - \frac{(n+1)(n-1)}{j+1}x^{\frac{1}{n+1}} + 1.$$

From here we can see that $f_{n,j}(x)$ is monotonic by noting that $n-2j+1 = (n-j)-(j+1)$. Therefore, when $x^{\frac{1}{n}} > j/n$ we can conclude that

$$I_{x^{\frac{1}{n}}}(n-j, j+1) - I_{x^{\frac{1}{n+1}}}(n-j+1, n-j) \geq 0.$$

Assume $x^{\frac{1}{n+1}} < j/n$:

We will show that

$$I_{x^{\frac{1}{n}}}(n-j,j+1) - I_{x^{\frac{1}{n+1}}}(n-j+1,j+1) > 0.$$

We can write

$$\begin{aligned} f_{n,j}(x) &= I_{x^{\frac{1}{n}}}(n-j,j+1) - I_{x^{\frac{1}{n+1}}}(n-j+1,j+1) \\ &= \int_0^{x^{\frac{1}{n}}} \frac{y^{n-j-1}(1-y)^j}{B(n-j,j+1)}dy - \int_0^{x^{\frac{1}{n+1}}} \frac{y^{n-j}(1-y)^j}{B(n-j+1,j+1)}dy. \end{aligned}$$

An application of the chain rule yields

$$\begin{aligned} f'_{n,j}(x) &= \frac{\left(x^{\frac{1}{n}}\right)^{n-j-1}\left(1-x^{\frac{1}{n}}\right)^j\left(x^{\frac{1-n}{n}}\right)}{nB(n-j,j+1)} - \frac{\left(x^{\frac{1}{n+1}}\right)^{n-j}\left(1-x^{\frac{1}{n+1}}\right)^j\left(x^{-\frac{n}{n+1}}\right)}{(n+1)B(n-j+1,j+1)} \\ &= \frac{\left(x^{\frac{1}{n}}\right)^{n-j-1}\left(1-x^{\frac{1}{n}}\right)^j\left(x^{\frac{1-n}{n}}\right)}{nB(n-j,j+1)} - \frac{\left(x^{\frac{1}{n+1}}\right)^{n-j}\left(1-x^{\frac{1}{n+1}}\right)^j\left(x^{-\frac{n}{n+1}}\right)}{(n-j)B(n-j,j+1)} \\ &= \frac{1}{B(n-j,j+1)}\left(\frac{\left(x^{\frac{1}{n}}\right)^{n-j-1}\left(1-x^{\frac{1}{n}}\right)^j\left(x^{\frac{1-n}{n}}\right)}{n} - \frac{\left(x^{\frac{1}{n+1}}\right)^{n-j}\left(1-x^{\frac{1}{n+1}}\right)^j\left(x^{-\frac{n}{n+1}}\right)}{n-j}\right) \end{aligned}$$

where we used $B(n-j+1,j+1) = \frac{n-j}{n+1}B(n-j,j+1)$ in the second equality. Now define $A_{n,j} = \{x : f'_{n,j}(x) \geq 0\}$. For $x \in A_{n,j}$ we have

$$\begin{aligned} &\frac{\left(x^{\frac{1}{n}}\right)^{n-j-1}\left(1-x^{\frac{1}{n}}\right)^j\left(x^{\frac{1-n}{n}}\right)}{n} \geq \frac{\left(x^{\frac{1}{n+1}}\right)^{n-j}\left(1-x^{\frac{1}{n+1}}\right)^j\left(x^{-\frac{n}{n+1}}\right)}{n-j} \\ \implies &\frac{\left(x^{\frac{1}{n}}\right)^{n-j-1}\left(1-x^{\frac{1}{n}}\right)^j\left(x^{\frac{1-n}{n}}\right)}{\left(x^{\frac{1}{n+1}}\right)^{n-j}\left(1-x^{\frac{1}{n+1}}\right)^j\left(x^{-\frac{n}{n+1}}\right)} \geq \frac{n}{n-j} \\ \implies &x^{-\frac{j}{n(n+1)}}\left(\frac{1-x^{\frac{1}{n}}}{1-x^{\frac{1}{n+1}}}\right)^j \geq \frac{n}{n-j} \\ \implies &\frac{x^{-\frac{1}{n(n+1)}} - x^{\frac{1}{n+1}}}{1-x^{\frac{1}{n+1}}} \geq \left(\frac{n}{n-j}\right)^{\frac{1}{j}}. \end{aligned}$$

Let

$$g_n(x) = \frac{x^{-\frac{1}{n(n+1)}} - x^{\frac{1}{n+1}}}{1-x^{\frac{1}{n+1}}}.$$

We can see that $g_n(x) \in (1,+\infty)$ for $x \in (0,1)$, $g_n(x)$ is decreasing, and $g_n(x) \to 1$ as $x \to 1$. Thus $A_{n,j} = (0, x^*_{n,j}]$ where $x^*_{n,j}$ is such that $g_n(x^*_{n,j}) = (\frac{n}{n-j})^j$. Putting it all together, $f_{n,j}(0) = f_{n,j}(1)$ implies that

$$I_{x^{\frac{1}{n}}}(n-j,j+1) - I_{x^{\frac{1}{n+1}}}(n-j+1,j+1) > 0$$

for all $x \in (0,1)$.